

1. Continuation & Refinement of Core Algorithms

Learning Velocity Algorithm v2.0+

- Document iterations beyond Attempt 14 (Dec 2024)
- Any new feature engineering approaches tested in 2025
- Cold-start problem solutions for new students with limited data
- Cross-subject calibration challenges (math vs. reading velocity)
- Age-normalization experiments (Grade 1 vs. Grade 12 learning patterns)

Teaching Velocity Model Enhancements

- Expanded TVV (Teaching Velocity Vector) beyond 20 features
- Real-time velocity computation challenges
- Multimodal fusion experiments (combining video, audio, transcript data)

2. New Technical Uncertainties for 2025

Privacy-Preserving AI Processing

- Federated learning experiments to process student data locally
- Differential privacy implementations for velocity calculations
- Edge computing for video analysis (processing on device vs. cloud)

Real-Time Session Analytics

- Low-latency behavioral signal processing during live sessions
- Stream processing architecture challenges
- Balancing accuracy vs. speed for in-session recommendations

Scalability & Performance Optimization

- GPU optimization for concurrent video analysis
- Database query optimization for velocity calculations at scale
- Caching strategy experiments for matching engine

3. Domain-Specific LLM Development

- Model architecture selection experiments (fine-tuning vs. training from scratch)
- Training data creation methodology (synthetic tutoring conversations?)
- Evaluation metrics for tutoring-specific responses
- Hallucination reduction techniques
- Privacy-safe training approaches
- Prompt engineering experiments for tutoring scenarios

4. AI Tutor Avatar - Move from Concept to Prototype

- Voice cloning experiments and quality assessment
- Teaching style transfer learning approaches
- Personality consistency across sessions
- Safety guardrails for AI-generated tutoring content
- Human-in-the-loop escalation triggers